

Aditya Bidwai

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Summary: Robotics Software Engineer with 2+ years deploying autonomy on real hardware across autonomous systems (UGVs, legged robots, and UAVs), specializing in motion planning, controls, and state estimation (C++, Python, ROS 2).

EDUCATION

University of Minnesota Twin Cities

Master of Science (M.S.), Robotics

Minneapolis, MN, USA

Sep 2024 – Jul 2026

Birla Institute of Technology and Science, Pilani (BITS-Pilani)

Bachelor of Engineering (B.E.), Electronics and Communication

Goa, India

Aug 2018 – May 2022

TECHNICAL SKILLS

Programming	C, C++ (Modern C++), Python, MATLAB, Bash
Tools & Frameworks	Git, Linux, Docker, CMake, Bazel, Robotics: ROS, ROS2, MoveIt!, PX4, RViz, Gazebo, Isaac Sim
Hardware	Raspberry Pi, NUC, Nvidia Jetsons, Arduino, ESP32, STM32, RealSense cameras, Optitrack MoCap
Domain Interests	Robotics Software, System Integration, Motion Planning, Trajectory Optimization, Optimal Control

EXPERIENCE

Komatsu Mining Corp

Robotics Software Intern (Automation)

Milwaukee, WI

Jun 2025 – Aug 2025

- Developed a startup motor/encoder calibration algorithm in C++ for robotic excavator to eliminate manual motor homing
- Built a C++ teleop interface (steering wheel/pedal) for a PX4 truck by reverse-engineering its RC unit via Arduino + DAC

University of Minnesota Twin Cities, **OptimalX Research Lab**

Research Assistant

Minneapolis, MN

Jan 2025 – Present

- Developed motion planning algorithms for UAV monitoring in energy-constrained environments (wildfires) using Mixed-Integer Linear Programming (MILP) -based optimization (Gurobi) and Markov Decision Process (MDP) methods
- Built an autonomous UAV (PX4, Pixhawk, RPi) for GPS-denied flight via ROS 2 offboard control and EKF2 mocap fusion
- Developed a robotics research lab from scratch (truss design, OptiTrack camera system procurement) serving 500 students
- Mentored undergraduate teams developing ground robot and Crazyflie platforms for department research projects

National University of Singapore, **MARMot Lab**

Research Engineer (Robotics)

Singapore

Jan 2022 – Aug 2024

- Deployed a novel gaze-planning (heading optimization) algorithm for legged robot mapping with fixed field-of-view cameras on a real hexapod (ROS, RTAB-Map, RealSense) ([paper](#))
- Co-developed an RL-based multi-robot task allocation method, 100x faster than optimization baselines ([ICRA '24 paper](#))
- Deployed an inverse kinematics-based controller for omnidirectional locomotion for a hexapod in Python/ROS ([video](#))
- Built a motion capture flight arena and a PX4 drone using ESP8266 offboard position control eliminating onboard computer

BITS Goa Mars Rover Team (Kratos), University Rover Challenge

Controls Team Lead

Goa, India

Aug 2020 – Aug 2021

- Led the controls subsystem, mentoring 10 undergraduate students in robotics software and system integration
- Designed path planning and tracking controllers (PID, Stanley, Pure Pursuit) for autonomous rover navigation
- Developed ROS/C++ drivers and software interfaces integrating sensors and motor drivers with an NVIDIA Jetson

SELECTED PROJECTS

Optimization-based Motion Planning for 3D Environments ([code](#))

- Built a convex optimization-based collision-free trajectory generation framework in MATLAB using YALMIP/MOSEK
- Modeled cuboidal and cylindrical obstacles as convex sets using superellipse-based formulations for tractable optimization

Cooperative UAV-UGV localization using Extended Kalman Filter (EKF) ([code](#))

- Developed a nonlinear UAV-UGV localization system in MATLAB using an EKF and simulation-based validation
- Evaluated estimator consistency using Monte Carlo testing with NEES/NIS analysis on simulated and real data

Dynamic Visual SLAM for Crowded Indoor Environments

- Integrated ORB-SLAM3 pipeline with Nvidia SegFormer to detect and reject moving objects in crowded environments
- Evaluated in real university environments, achieving 0.077 m ATE and 0.073 m / 1.33° RPE, showing robust localization

Clothbot - Cloth Manipulation using Self Supervised Value Network ([poster](#), [web](#))

- Developed a self-supervised value network policy using spatial action maps for dynamic cloth unfolding on a dual UR5
- Achieved 95% coverage on rectangular cloths and 87.68% on unseen garments (T-shirts) with zero-shot sim-to-real transfer

Flying Ad Hoc Network Simulator for multi-UAV exploration ([code](#))

- Developed a co-simulation platform integrating NS3 and Gazebo through ROS/C++ for testing multi-UAV swarm tasks
- Implemented UAV swarm motion planning (C++) and analyzed network metrics like PDR, hop-by-hop and end-to-end delay
- Simulated a wildfire rescue UAV swarm (PX4 SITL and ROS) for environment surveillance applications ([ACM paper](#))